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SUPERSEDES
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MILITARY SPECIFICATION

CONVERTER, SINGLE SIDEBAND, CV-157( )/URR

1. SCOPE

1.1 This specification covers the requirements for construction of Converter, Single Sideband CV-157( )/URR. This equipment is capable of receiving single or double sideband signals from a radio receiver whose IF frequency is in the range of 450 to 510 KC. It separates the sidebands and converts them into audio signals for operation of terminal equipment or for distribution to land lines or other radio circuits. The equipment is capable of operation from either a nominal 115 volt, or a 230 volt, 50–60 cycle, single phase power source.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

T-T-871 — Twine, Cotton, Wrapping.

QQ-S-571 — Solder; Lead Alloy, Tin Lead Alloy, and tin Alloy; Flux Cored Ribbon and Wire, and Solid Form.

LLL-F-291 — Fiberboard; Corrugated, Single Face (flexible).

PPP-B-585 — Boxes; Wood, Wirebound.

PPP-B-591 — Boxes; Fiberboard, Wood-Cleated.

PPP-B-621 — Boxes, Wood, Nailed and Lock-Corner.

PPP-B-636 — Boxes. Fiber.

PPP-T-76 — Tape, Pressure Sensitive Adhesive, Paper, Water Resistant.

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MIL-P-75 — Packaging, Packing and Container-Marking of Electron Tubes.

MIL-P-116 — Preservation, Methods of.

MIL-B-117 — Packaging and Packing for Overseas Shipment, Bags, Interior Packaging.

MIL-T-152 — Treatment, Moisture-and-Fungus-Resistant, of Communications, Electronic,

FSC 5820

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and Associated Electrical Equipment.

MIL–T–713 — Twine and Tape, Lacing and Tying (for use in Electrical and Electronic Equipment).

MIL–I–7798 — Insulation Tape, Electrical, Pressure-Sensitive Adhesive, plastic.

MIL–P–11268 — Parts, Materials, and processes used in Electronic-Communication Equipment.

MIL–M–13231 — Marking of Electronic Items.

MIL–T–14015 — Tape, Nylon, Waxed.

MIL–F–14072 — Finishes for Ground Signal Equipment.

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

MIL–STD–169 — Extreme-temperature cycle.

MIL–STD–170 — Moisture resistance test cycle for ground signal equipment.

MIL–STD–252 — Wired Equipment, Classification of Visual and Mechanical Defects.

DRAWINGS

SC–DL–180165 — Converter, Single Sideband CV–157 ( )/URR.

Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Uniform Freight Classification No. 4
Uniform Freight Classification
202 Union Station
Chicago 6, Illinois

Consolidated Freight Classification Rules No. 21
Consolidated Classification Committee
1 Park Avenue and 33rd Street
New York 16, New York

National Motor Freight Classification No. A–3 and No. 14
American Trucking Association, Inc.
1424 Sixteenth Street
Washington 6, D. C.

3. REQUIREMENTS

3.1 Description. The converter is intended for use in applications between fixed installations having a heavy flow of message traffic. The output signal of the converter will be identical with the multiplexed signals delivered over the wires in voice-frequency landline carrier telephony. Thus, the converter. operating with a radio receiver, may be used to feed carrier terminal equipment that operates several channels of teletype-writers, facsimile, or voice reproducing equipment.

3.2 Construction. Converter, Single Sideband CV–157( )/URR shall be constructed in accordance with Drawing and Data List SC–DL–180165.

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3.3 Parts, materials and processes; general. In addition to the requirements of this specification, the requirements of MIL–P–11268 including the selection requirements therein, shall apply.

3.3.1 *Wiring and cabling.* Wiring and cabling shall be neat and sturdy.

3.3.1.1 *Cabling.* Insulated wires shall be formed into cables except where operation of the equipment would be adversely affected thereby or where it is physically impracticable as in the case where the resulting cables would be excessively large and would interfere with operation or maintenance of the equipment. The cabling of wires shall be effected by twine conforming to MIL–T–713 or tape conforming to Specification MIL–I–7798 or MIL–T–14015.

3.3.1.2 *Slack.* Wires and cables shall be as short as practicable except that sufficient slack shall be provided to prevent undue stress on cable forms, wires, and connections; to enable parts to be removed and replaced during servicing without disconnecting other parts; and to facilitate field repair of broken or cut wires.

3.3.1.3 *Protection.* Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material. Where wires run through holes or slots in metal (partitions, shields, and similar items), they shall be protected by suitable grommets or bushings.

3.3.1.4 *Support.* Wires and cables shall be properly supported and secured, to prevent undue stress on conductors and terminals and undue change in position of the wires and cables after the equipment has been subjected to specified service conditions or has been serviced or repaired in a normal manner.

3.3.1.5 *Clearance.* Clearance between wires or cables, and parts such as electron tubes, resistors, and dynamotors, shall be sufficient to avoid deterioration of the wires or cables because of the heat dissipated by such parts when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards, relays, or other parts, shall be at least ¼ inch unless use of specified parts makes such clearance impracticable.

3.3.1.6 *Splicing.* Wires in a continuous run between two terminals shall not be spliced during assembly of equipment.

3.4 Soldering.

3.4.1 *Solder.* Solder used for electrical connections shall be composition Sn60 Conforming to Specification QQ–S–571.

3.4.2 *Flux and cleaning agents.* No acid or acid salts shall be used in preparation for or during soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, ts permitted above, they shall be completely neutralized and removed immediately after use. Flux for soldering of electrical connections shall be rosin, rosin and alcohol, or rosin and turpentine.

3.4.3 *Process.* There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting action, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment;

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flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.5 Cleaning. Metal parts, after fabrication, shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect on the equipment. Corrosive material shall be removed completely before parts are assembled into the equipment. After assembly components shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. If necessary, such cleaning shall also be performed before final assembly of components.

3.6 Finish, protective. Equipment shall be given protective finish in accordance with MIL–F–14072. This includes finish of hardware, such as handles, hinges, screws, etc., and necessary touch up after mounting. The final paint film on type 1 surfaces shall be final film E, semigloss, light-gray enamel, conforming to MIL–F–14072.

3.7 Marking.

3.7.1 *Conformance.* Marking shall conform to MIL–M–13231. Front panel marking shall be group 1.

3.7.2 *Visibility.* Wherever practicable, parts and assemblies shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.7.3 *Serial numbers.* All converters shall be serial numbered.

3.8 Securing of parts.

3.8.1 *General.* Brackets, lugs, flanges, inserts, bolts and other mounting arrangements shall be such as to retain components and parts securely when the equipment is subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the equipment is subjected to specified bounce and shock tests. Any permanent deformation resulting from the tests shall not cause degradation of equipment performance below full specification requirements. Where practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely.

3.9 Tropicalization.

3.9.1 *Treatment of assembled equipment.* Assembled equipment shall be treated in accordance with MIL–T–152 unless both of the following conditions are met:

(a) Each component part and subassembly meets one of the following:

(1) It is furnished under an individual, detailed Government document (specification, standard, drawing, etc.) which is applicable on contract.

(2) It does not incorporate any material listed in 3.9.7 unless the material has been given the treatment specified herein or a treatment approved by the contracting officer.

(3) It does not comply with (1) or (2) above but its use has been approved by the contracting officer for the particular application.

(b) The finish of the interior of the equipment meets the requirements of MIL–F–14072.

3.9.2 *Treatment of materials.*

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3.9.3 *Treating materials.* Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material is held to the practical minimum.

3.9.4 *Toxicity.* Treatment of materials shall cause no skin irritation or other injury to personnel handling the treated material either during fabrication of the equipment or when carrying, operating, or maintaining the equipment, or in use of the finished items when used for the purpose intended.

3.9.5 *Flexibility.* Treatment shall not affect flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.9.6 *Statement of treatment.* The contractor shall submit, to the contracting officer for approval, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. When assembled equipment is treated in accordance with MIL–T–152, the above statement shall be incorporated in the statement of treatment required by that specification.

3.9.7 *Materials not resistant to moisture and fungi.* The following materials are considered not resistant to moisture and fungi:

(a) Cellulose, regenerated.

(b) Cotton.

(c) Cork.

(d) Felt, hair or wool.

(e) Jute.

(f) Leather.

(g) Linen.

(h) Paper, paperboard, cardboard, organic fiberboard, vulcanized fiber, etc.

(i) Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base.

3.10 Electrical requirements.

3.10.1 *Operation.* Single Sideband Converter CV–157( )/URR shall be capable of operation with any radio receiver whose IF frequency lies in the range of 450 to 510 kc, and whose IF bandwidth, between minus 1 db points, is not less than 12 kc. The input impedance shall be between 47.5 and 52.5 ohms resistive.

3.10.2 *Local and reconditioned carrier.* Single Sideband Converter CV–157( )/URR shall provide both a locally generated and a reconditioned carrier. Local carrier voltage shall be derived from a crystal controlled oscillator operating at the center frequency of the converter IF bandpass filter. The reconditioned carrier shall be obtained by passing the IF signal through a narrow band filter to remove sideband voltages, and by limiting it to remove variations in its amplitude.

3.11 Indicators.

3.11.1 *Drift indicator.* The drift indicator shall be calibrated in KC to measure the AFC correction for drift of the carrier input to Single Sideband Converter CV–157( )/URR, and shall be accurate within ±200 cps. The drift indicator alarm shall be adjusted to give an indication of at least 300 cps before the system loses control of the mixing oscillator on each side of the drift extremes (see 4.30).

3.11.2 *Carrier level meter.* The carrier level meter shall indicate carrier level, and shall be calibrated to an arbitrary reference using a linear scale. The normal carrier level for satisfactory carrier operation of Single Sideband Converter CV–157( ) URR shall

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be designated with the number 10 at approximately 50 percent of full scale deflection, and other scale deflections shall be labeled accordingly. When tested in accordance with 4.13, the readings of the carrier lever meter shall be between 17 to 20 plus, 6.5 to 7.5, 4.5 to 5.5, 2.0 to 3.0 and 0.7 to 1.7 respectively.

3.11.3 *VU meter.* A standard volume indicator (VU meter) shall be used to measure the level of the audio signal on each output line to within —1.0 db of the true value. The switch shall transfer the meter between the two sideband outputs, or disable the meter completely. The attenuator pad in the measuring circuit shall extend the range of the VU meter 10 db, above and below its normal range. The volume indicator circuit shall be so calibrated, that the algebraic sum of the meter reading and the attenuator switch setting shall indicate the true volume level at the output terminals of Single Sideband Converter CV-157( )/URR. Operation of the VU meter and associated pads and switches shall not change or effect levels being fed to output lines (see 4.15).

3.11.4 *Audio output circuits.* Adjustment of the output level and loading of one audio output shall not affect the level of the other audio output circuit. Each output circuit shall be capable of providing up to 100 milliwats power output for 600 ohms line or phones with a signal input of 10,000 microvolts SSB, or with an input of 10,000 microvolts DSB signals using 30 percent amplitude modulation (see 4.16).

3.11.5 *Monitoring circuit.* The monitoring circuit shall deliver a minimum of 100 milliwatts into a 600 ohm load, when 1 milliwatt is delivered to each line properly loaded. The switch on the front panel shall be capable of selecting either the upper side band or the lower sideband. Operation of this circuit shall not effect any of the audio output circuits of the converter in any way. The monitor gain control shall be capable of adjusting the level of the monitored signal (see 4.31).

3.12 Selectivity.

3.12.1 *Carrier rejection in sideband channels.* Sideband filters shall have not less than 10 db rejection at the IF carrier frequency. Rejection is defined as a measure of the difference of insertion loss at 100 kc and the average of the maximum and minimum insertion loss across the pass band (see 4.17).

3.12.2 *Carrier Band—Pass Filter.* The bandwidth VS attenuation requirements for the carrier band-pass filter shall be as follows:

Attenuation	*Bandwidth*
1 db	19 cps min
55 db	160 cps max
60 db	180 cps max

In the range carrier-frequency plus 130 cps to plus 6000 cps and carrier-frequency minus 130 cps to minus 6000 cps, the attenuation shall be at least 70 db. Beyond the range carrier frequency plus and minus 6000 cps, the attenuation shall remain at least 70 db (see 4.18).

3.12.3 *Rejection of unwanted sideband.* When a test signal with an upper sideband voltage of 10,000 microvolts and a carrier voltage of 1,000 microvolts is impressed on the input terminals of Single Sideband Converter CV-157( )/URR, and with the audio output level of Channel A and Channel B equalized, for sideband frequencies in the range 200 cps to 6000 cps above the carrier frequency, the level of the output signal of the lower sideband shall be not less than 55 db below that of the upper sideband. A sideband signal in the frequency range 100 cps to 199 cps above the carrier frequency shall be attenuated at least 42 db in the lower sideband. The same discrimination shall be provided when the test signal is applied to the lower sideband, and the voltages measurement is made on the upper sideband (see 4.19).

3.13 Noise and distortion.

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3.13.1 *Total distortion products.*

3.13.2 *Single sideband.* When a composite test signal consisting of a carrier frequency voltage of 1000 microvolts, and two sideband voltages of 5000 microvolts each, differing in frequency from the carrier by between 100 and 6000 cps, and from each other by not more than 500 cps, is applied to the input terminals of Single Sideband Converter CV-157( )/URR, any resulting distortion component in the output signal shall be not less than 42 db below the output level of one of the two equal sideband voltages, when the output level is set at 10 milliwatts (see 4.21).

3.13.3 *Overall audio frequency response.* The following shall apply to both upper and lower sideband signals. When a single tone SSB of constant amplitude of 10,000 microvolts, together with a carrier of 1000 microvolts is impressed on the input terminals of Single Sideband Converter CV-157( )/URR, the difference between maximum and minimum response, measured across a 600 ohm load connected to the converter output terminals, shall be not greater than 2 db for frequencies in the band between 125 and 6000 cps. The output level shall drop smoothly without reverse curvature on each side of the 100 to 6000 cps pass band (see 4.22).

3.13.4 *Hum level.* With a 10,000 microvolt input signal modulated 30 percent at 400 cps, and the audio output adjusted for maximum output, the hum output shall be at least 50 db below the 400 cps audio output signal. When a composite test signal consisting of two sideband voltages of 5000 microvolts each, differing in frequency from the carrier by between 100 and 6000 CPS, and from each other by not more than 500 CPS, is applied to the input terminals of Single Sideband Converter CV-157( )/URR, and with the audio output level set at 10 milliwatts, the hum shall be at least 50 db below a single test signal (see 4.20).

3.14 Automatic frequency control system.

3.14.1 *AFC Pull-in-Range.* The AFC system shall have a pull-in range of plus-or-minus 50 cps from the center frequency of the carrier filter with the SQUELCH circuit off. It shall hold-in over a range of plus-or-minus 2000 cps from the center frequency of the carrier filter (see 4.23).

3.14.2 *AFC drift rate.* The AFC system, when locked on an incoming carrier, shall be capable of following an overall system frequency drift of up to 10 cycles per second (see 4.24).

3.14.3 *Residual AFC error.* The maximum allowable residual AFC error shall be plus or minus 2 cycles/sec average or 2½ cycles/sec PEAK (see 4.25).

3.14.4 *Mixer Oscillator Stability.* The mixing oscillator shall show a drift of no greater than 100 cps when measured an hour after the three minute cold start. During the two minute measurement after the unit has warmed up, the mixing oscillator frequency shall show less than 10 cps drift for any one minute period. The instantaneous drift rate shall be less than 5 cps at any point (see 4.14).

3.14.5 *Carrier-noise squelch circuit.* The squelch circuit shall stabilize the mixing oscillator frequency when there is a decrease in the carrier signal quality. This circuit shall go into operation when the carrier signal-to-noise power ratio drops below the pre-adjusted operating level. This protective circuitry shall function when the carrier drops below the pre-set value; when the noise level increases above a preset value, or when both conditions occur simultaneously. The operating point shall only occur when two carrier signals, one at normal level (half scale), and the other 3 db lower but 5 cycles removed in frequency are applied simultaneously to the carrier channel of the converter (see 4.26).

3.14.6 *Automatic gain control characteristics.* Single sideband Converter CV-157

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( )/URR shall develop an AGC voltage in accordance with variations of the selected signal component. The relationship between the AGC output voltage and the IF input voltage, when the converter is used with Radio Receiver R–390( )/URR, shall result in a constant audio output from the converter. The overall system gain shall be controlled so that the audio output from the converter shall not vary, in each of the SB select switch positions, by more than 6 db for a given input to the radio receiver. The input to the receiver shall be referenced to the sideband signal which shall vary from 50 microvolts to 1.0 volt. The carrier signal shall be maintained at a level of 20 db below that of the sideband signal. The time constant of this circuit shall be adjustable in three fixed steps. The first shall be determined by the minimum time constant of the radio receiver with which the converter is used. The second step shall cause the overall time constant to be not less than one second. The third shall provide for a time constant of fifteen seconds (see 4.28).

3.14.7 *Carrier level compensation.* The filtered carrier circuit of Single Sideband Converter CV–157( )/URR shall introduce continuously adjustable gain under the control of the front panel knob calibrated in db. The circuit shall provide a smooth, continuous variation of at least 30 db (see 4.29).

3.14.8 *Low pass filters.* The low-pass filters shall have a cutoff at 3500 cps in each channel. An internal switch shall be used to connect the filters into their respective circuits. With the filters connected, the response of the channel shall be as specified in 3.13.3, except that the upper limit of frequency shall be 3500 cps. Energy in the frequency range 4200 to 6000 cps shall be attenuated by not less than 55 db with no rise in output beyond 6000 cps (see 4.22).

3.15 Power supply.

3.15.1 *Power supply noise.* Acoustic noise from the power supply shall be kept to a minimum. Evidence of lamination vibration in transformers or chokes shall be cause for rejection.

3.15.2 *Power Source.* The converter shall operate from a nominal 115 volt or 230 volt, 50–60 cycles per second, single phase supply. The circuit breakers, shall not require adjustment to compensate for any temperature change within the requirements of this specification, and shall be direct acting from off to on and vice versa. They shall provide visual indication of their position, either off or on.

3.15.3 *Range of power source variation.* Single Sideband Converter CV–157( )/URR shall operate when a single phase potential difference of 100 to 130 volts, is maintained across the power input terminals, when the voltage selector switch is set to the 115 volt position. The potential difference range shall be 200 to 260 volts in the 230 volt position. Power Supply frequencies of 48–62 cps shall be accommodated. When any supply voltage and frequency within these ranges is in use, the converter shall operate under the service conditions as specified by 3.18. The equipment shall meet all the requirements of this specification when any supply voltage between 110 and 120 volts rms, or between 220 and 240 volts rms, as applicable, is applied.

3.16 Duty cycle. Single Sideband Converter CV–157( )/URR shall be capable of meeting the test of 4.12. An inspection shall be made at the end of the period for evidence of deterioration and malfunction resulting from the extended continuous duty.

3.17 Quartz crystal filters.

3.17.1 *Filter requirements.* The filters shall be tested prior to assembly in the equipment in accordance with 4.27 and shall meet the applicable requirements listed below:

The lower sideband filter shall be flat

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within 1 db from 94.0 – 99.875 kc. The upper sideband filter shall be flat within 1 db from 100.125 – 160.0 kc. The maximum insertion loss for these filters shall not exceed 7.0 db. Spurious notches (spikes), shall not exceed 2 db across the pass band of the sideband filters.

Carrier filter				
Test #	Freq KC	Insertion Min.	Loss DB Max.	
1	100.000	—	6.0	
2	100.005	M–0.3	M+0.7	
3	99.995	M–0.3	M+0.7	M = Actual loss measured in Test #1
4	100.030	M+15.5	—	
5	99.970	M+15.5	—	
6	100.090	58	—	
7	99.910	58	—	

3.18 Service conditions. The equipment shall meet the following service conditions:

3.18.1 ***Equipment nonoperating.*** Except as stated in 3.18.1(b), the equipment shall comply with the operational requirements of 3.18.2 after subjection to any of the following nonoperating conditions successively or in combinations encountered during worldwide short-term storage and transit:

(a) *Temperature.* Exposure in the range of +160°F to —80°F; however, exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 24 hours at any one time.

(b) *Relative humidity.* Relative humidity up to 97 percent, for an indefinite period of time; and exposure at 100 percent relative humidity with condensation, for 4 hours. Following subjection to such conditions and a power-on period of 16 hours at normal room temperature and relative humidity, the equipment shall meet the requirements of 3.18.2.

(c) *Elevation.* Elevation up to 25,000 feet above sea level.

(d) *Vibration, bounce, and shock.* As described in 3.19.

3.18.2 ***Equipment operating.*** After a warmup period of ½ hour, the equipment shall meet the performance requirements of this specification while subjected to any of the following conditions successively or in combinations likely to be encountered during worldwide operations:

(a) *Operation.* Continuous operation for a period of 1 year, alternating 24 hours on and 0 hours off, with no more than normal maintenance and replacement of parts. Compliance will be determined by examination of the preproduction sample.

(b) *Temperature.* Ambient temperature in the range of +150°F to —65°F. (The 150°F temperature includes effect of sun-load.) Exposure at the high temperature extreme not to exceed 4 hours and at the low temperature extreme not to exceed 72 hours.

(c) *Relative humidity.* Relative humidity up to 97 percent; and exposure at 100 percent relative humidity, with condensation, for 4 hours.

(d) *Elevation.* Elevation up to 10,000

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feet above sea level.

3.19 Vibration, bounce, and shock.

3.19.1 *Vibration, internal.* The amplitude of vibration of any part, subassembly, or structural member of the converter, shall not exceed twice the amplitude of the vibration applied to the converter at any frequency between 10 and 55 cycles per second (see 4.8).

3.19.2 *Bounce and shock.* The equipment shall meet the requirements of table I.

TABLE I. *Bounce and shock*

Requirement	Inspection paragraph	Performance after inspection
Bounce, light equipment	4.10	Full specification performance. No physical damage except surface abrasion.
Shock; bench-handling	4.6	Full specification performance. No physical damage.

3.20 Interchangeability. Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. Individual items shall not be hand-picked for fit or performance; however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc. (see 4.33).

3.21 Preconditioning. Items shall be capable of meeting the inspection of section 4, without subsequent processing, after subjection to the bounce preconditioning of 4.4.6 (also see 4.4).

3.22 Preproduction samples. The contractor shall furnish preproduction samples for approval, as required by the invitation for bids and contract.

3.23 Technical literature, tools, and running spare pares. Technical literature, tools, and running spare parts shall be furnished as specified in the contract. Running spare parts shall be identical to corresponding parts in the items furnished on the order.

3.24 Workmanship. The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs:

(a) In Specification MIL–P–11268:

Fabric and thread
Ceramic material and parts
Lubricants
Fasteners for access devices

(b) In this Specification:

3.3.1 Wiring and cabling
3.4 Soldering
3.5 Cleaning
3.8 Securing of parts
3.9 Tropicalization

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.2 *Classification of inspection.* Inspection shall be classified as follows: (See 6.4)

(a) Preproduction inspection (does not include preparation for delivery) (see 4.2).

(b) Inspection covered by subsidiary documents (see 4.3).

(c) Quality conformance inspections.

(1) Quality conformance inspection of equipment before preparation for delivery (see 4.4).

(2) Quality conformance inspection of preparation for delivery (see 4.34 and Section 5).

4.2 Preproduction inspection. This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table II, inspeciton specified in the subsidiary documents covering the items listed in 4.3, and the group A, group B, and group C inspection specified in tables III, IV, and V, respectively. Other nondestructive tests on preproduction samples may be performed to determine compliance with specified requirements. The preproduction inspection will normally be performed in this order: (1) vibration, (2) bounce, and (3) shock, benchhandling; other preproduction inspection may precede, follow, or be interspersed between the foregoing.

TABLE II. *Preproduction inspection*

Inspection (For additional preproduction inspection see 4.2)	Requirement paragraph	Inspection paragraph
Vibration	3.19.1	4.8
Temperature	3.18.1(a)	4.7
Bounce	3.19.2	4.10
Shock; Benchhandling	3.19.2	4.6
Relative Humidity	3.18.1(b)	4.5
Elevation	3.18.1(b)	4.32

4.3 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection before preparation for delivery.

Item	*Where required*
Finish	3.6
Marking	3.7

4.4 **Quality Conformance inspection of equipment before preparation for delivery.** The contractor shall perform the inspection specified in 4.3 and 4.4 through 4.4.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government—at its discretion—may perform all or any part of the specified inspection to verify the contractor's compliance with specified requirements. (See 6.5). Test equipment for Government verification inspection shall be made available by the contractor. Each unit which shall be subjected to group A, group B and/or group C inspection, except preproduction samples, shall be bounce preconditioned (see 3.21) after final assembly.

4.4.1 *Group A inspection.* This inspection, including sampling, shall conform to table III and the general inspection procedure of MIL–STD–105. Group A inspection shall be performed in any order which is satisfactory to the Government.

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TABLE III. *Group A inspection*

Inspection	Requirement paragraph	Inspection paragraph	AQL	
			Major	Minor
Visual and Mechanical:				
Drawer Assembly	3.24	4.9	10 dphu *	25 dphu
Base Assembly	3.24	4.9	2.5 dphu	10 dphu
Complete Assembly	3.24	4.9	1.0 dphu	4.0 dphu
Electrical: *				
Residual Error	3.14.3	4.25)		
Carrier Noise Squelch Circuit	3.14.5	4.26)		
Carrier Lever Meter	3.11.2	4.13)		
Drift Indicator	3.11.1	4.30)	6.5 dphu	
VU Meter	3.11.3	4.15)	for	
Carrier Rejection	3.12.1	4.17)	Group	
Carrier Band Pass Filter	3.12.2	4.18)		
Rejection of Unwanted Sideband	3.12.3	4.19)		
Single Sideband Distortion	3.13.2	4.21)		
Overall Audio Frequency Response	3.13.3	4.22)		
Pull-in Range	3.14.1	4.23)		
AFC Drift Range	3.14.2	4.24)		
Monitoring Circuit	3.11.5	4.31)		
Filter	3.17	4.27)	1.0%	

* dphu — Defects per hundred units.
** all electrical defects are considered major.

4.4.2 *Group B inspection.* This inspection, including sampling, shall conform to table IV and to the special procedures for small-sample inspection of MIL–STD–105. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection.

4.4.2.1 *Group B sampling plan.* The acceptable quality level (AQL) shall be 6.5% defective, as specified in table IV, and the inspection level shall be S–4 for normal and S–3 for reduced inspection.

4.4.2.2 *Order of inspection within Group B.* Group B inspection shall be performed in any order which is satisfactory to the Government.

TABLE IV. *Group B inspection*

Inspection	Requirement paragraph	Inspection paragraph	Group B plans (See 4.4.2.1)
Hum Level	3.13.4	4.20	
Mixing Oscillator Stability	3.14.4	4.14	
Interchangeability	3.20	4.33	6.5%
Low Pass Filters	3.14.8	4.22	For the
Audio Output Circuit	3.11.4	4.16	group
AGC Characteristics	3.14.6	4.28	
Carrier Level Compensation	3.14.7	4.29	

4.4.3 *Group C inspection.* This inspection shall be as listed in table V.

4.4.3.1 *Sampling for inspection of equipment.* Two equipments for each group C inspection shall be selected each month, without regard to their quality.

4.4.3.2 *Noncompliance.* If a sample unit fails group C inspection, the contractor shall

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immediately investigate the cause of failure and shall report to the Government inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer.

TABLE V. *Group C inspection*

Inspection	Requirement paragraph	Inspection paragraph
Continuous Duty	3.16	4.12
Power Cycle	3.15	4.35

4.4.4 ***Reinspection of conforming group B and C sample units.*** **Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage.**

4.4.5 *Disposition of nonforming product.* The following shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault:

Sample units found defective during group A, B, or C inspection.

Inspection lots which failed during group A or B inspection.

Production which has been reworked as a result of failure during group C inspection.

The required information shall be presented to the Government when the product is submitted and shall become the property of the Government.

4.4.6 *Bounce preconditioning.* The unit shall be placed in its normal operation position on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a $\frac{1}{32}$-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned, or otherwise changed prior to subjection to procurement inspection (see 3.21).

4.5 Moisture-resistance test for equipment. **The equipment shall be tested as follows:**

4.5.1 *Test conditions.*

(a) Do not move equipment from the humidity chamber for measurements.

(b) Start measurements not more than 5 minutes after power is applied to the equipment. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed equipment with the seal intact except during last two cycles. Bring out leads for electrical measurements through a hole in the equipment enclosure and seal against entrance of moisture. Perform last two cycles with the chassis removed from its enclosure.

4.5.2 *Test Method.*

(a) Dry at 130° ±5°F for 24 hours.

(b) Condition at 77°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.5.3, and readjust or realign as necessary to meet full specification requirements.

(d) Subject to continuous cycling for 5 48 hour cycles. Temperature, relative humidity, and period of

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time for each portion of the cycle shall conform to MIL-STD-170. The measurements specified in 4.13.3 shall be made during the periods shown on the standard and shall comply with 4.5.3.

(e) After cycling has been completed, condition the equipment for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided by the equipment. No repair or replacement of parts shall be made. After adjustment, the equipment shall meet full specified performance for those measurements specified in 4.5.3.

4.5.3 *Performance requirements.* There shall be no degradation in performance from that obtained prior to the 5 humidity cycles of 4.5.2(d).

4.5.4 *Failure.* If any equipment fails to meet the performance specified in 4.5.3 during cycling, it shall be realigned or readjusted once. If the equipment then fails to meet 4.5.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.5.2(e), it does not pass the test.

4.6 Shock test; bench-handling. The chassis-and-front-panel assembly shall be removed from its enclosure, as for servicing, and placed in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows, in a manner simulating shocks liable to occur during servicing, to determine conformance to 3.19.2:

(a) Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

(b) Repeat (a), with the assembly resting on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.7 Temperature. The equipment shall be subjected to the temperature cycle shown in MIL-STD-169. The equipment shall meet full specification requirements when tested at steps 1, 4, 5, 8, and 10 (see 3.18.1(a)).

4.8 Test for internal vibration. Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.19.1:

(a) Secure the converter directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, subassemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

(b) Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

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4.9 Visual and mechanical inspection. Equipment shall be examined for the defects listed in MIL–STD–252.

4.10 Bounce test; equipment under 1000 pounds. The equipment shall be placed on the table of the Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal, and shall be constrained from horizontal motion of more than 2 inches by suitable wooden fences. The package tester, shafts in phase, shall be operated at a speed of 285 rpm ± 1 percent, for a total of 3 hours. During each ½-hour period of the test, the transit case (with contained equipment) shall rest on a different face (see 3.19.2).

4.11 Standard test conditions. Unless otherwise specified, all performance tests shall be conducted under the following conditions:

4.11.1 *Temperature and humidity.* The standard test temperature and humidity shall be considered as prevailing room ambients.

4.11.2 *Supply voltage and frequency.* The standard test supply voltage and frequency shall be 115 volts, 60 cps.

4.11.3 *Signal Generation and controls.* Single and twin sideband test signals at 450 to 510 kc may be generated by three oscillators with a suitable combining circuit in place of a single sideband signal generator. For each test, unless otherwise specified, the following switches and controls shall be in the positions listed below:

Switch or Control	Position
Carrier Noise Squelch	Off
AFC	On
Carrier level compensation	20 db
Local reconditioned carrier	LC

4.11.4 *Input Gain adjustment.* Adjustment of the input gain control is necessary in order to provide the converter circuits with the proper levels required for the electrical tests. The input gain control shall be adjusted in the following manner: A single tone single sideband signal of constant amplitude of 10,000 microvolts, 2500 cycles from the IF center frequency, together with a carrier frequency of 1000 microvolts shall be impressed on the input terminals Single Sideband Converter CV–157( )/URR. The input gain control shall be adjusted to give a reading of 0.09 volts A.C. measured at terminal 3 of transformer T4. Once this adjustment is made the input gain potentiometer shall be locked, and no further adjustment of the input gain potentiometer shall be made.

4.12 Test of continuous duty cycle operation. Single Sideband Converter CV–157 ( )/URR shall be operated under standard conditions for a period of 30 hours without shutdown. An inspection shall be made at the end of the period for evidence of deterioration or malfunction resulting from the extended continuous duty (see 3.16).

4.13 Carrier level meter. A single generator set at 455 KC shall be connected to the input terminals of Single Sideband Converter CV–157( )/URR and centered in the carrier channel. The level of the signal generator shall be adjusted so that the carrier level meter reads half scale. The input signal shall then be changed by plus 6 db then minus 3 db, 6db, 12 db and 18 db. The corresponding readings of the carrier level meter shall be between 17 to 20 plus, 6.5 to 7.5, 4.5 to 5.5, 2.0 to 3.0 and 0.7 to 1.7 (see 3.11.2).

4.14 Mixing oscillator stability. The input and mixing oscillator circuits shall be tuned to 455 KC. Single Sideband Converter CV–157 shall be disconnected from the 115 volt (or 230 volt) power source for a period of at least two (2) hours. During this period a signal generator at a frequency of 455 KC, set for 10,000 microvolts. shall be in operation in order to stabilize its output voltage and frequency. When the converter has stabilized at room ambient temperature, the signal generator shall be connected to the

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converter input terminals. The converter shall then be turned on and the signal generator adjusted to lock the AFC to the incoming carrier frequency. After a three minute start, the frequency of the mixing oscillator shall be measured. After an hour's wait, during which the temperature variation becomes stabilized, readings shall be taken every two seconds for a period of two minutes. The requirements of par. 3.14.4 shall be met.

4.15 VU meter. The following conditions shall be established:

(a) A 1000 cps, 30 per cent amplitude modulated signal of 10,000 microvolts rms shall be applied to the input terminals of the converter.

(b) A general Radio Type 583–A Power Output Meter, or equal, adjusted for 600 ohms output, shall be connected between the applicable output terminals.

(c) The VU meter switch shall be switched to the zero position for the output channel under test.

(d) The volume control for the channel under test shall be adjusted so that the output meter reads 1 milliwatt.

The reading of the VU Meter shall be noted. The VU Meter switch shall be switched to the —10 db position and the volume level reduced until the output meter reads 0.1 milliwattt. The reading of the VU Meter shall be noted. The VU Meter switch shall then be switched to the + 10 db position and the volume level increased until the output meter reads 10 milliwatts. The reading of the VU Meter shall again be noted. The extent to which the VU Meter fails to indicate 0.0 VU for all tests shall be noted to determine compliance with the requirements of 3.11.3.

4.16 Audio output circuits. With the carrier level compensation control set at 20 db, a test signal consisting of a 1,000 microvolt carrier and two sideband signals, each of 5,000 microvolts, one 400 cps above the carrier frequency and the other 700 cps below shall be applied to the input terminal of Converter, Single Sideband CV–157( )/URR. A power output meter set to 600 ohms shall be connected to the line terminals of one of the channels, and its power output shall be set at 500 microwatts. The volume control of the other channel shall be varied over its entire range, and then with the volume full on, the load impedance on this channel shall be varied between 500 and 700 ohms. These tests shall not produce any perceptible variation in the output power developed by the other channel. This test shall be repeated with the variation of controls for the two channels interchanged (see 3.11.4).

4.17 Carrier rejection. Tests shall be made to insure compliance with the requirements of 3.12.1 by connecting an RF VT VM to pin 2 of S1, with the SB selector switch in the A–LSB position, record the db level indicated on VT VM.

4.18 Carrier band-pass filter. Connect an RF VT VM (H.P. 400 C or equal) with a shielded input to pin 2 of V19. Rock the signal generator frequency until a peak deflection is obtained on the VT VM. This is the center frequency, fc, of the Carrier Filter. Record this value as read on a Frequency Counter. All frequency readings are read from the Counter. Adjust the Carrier Compensator so that the VT VM reads some convenient value around +10 DB. (Do not disturb this setting for the remainder of the test.) Record this value. Disconnect the Carrier Filter, FL–5, and feed straight through. Record the output. The difference between the two readings gives the insertion loss. Reinsert the Filter. The signal generator frequency shall be varied above and below fc and the frequencies recorded for the 1 DB, 55 DB and 60 DB points. Outside the passband the input signal attenuation should be reduced as required. Work with as strong an input signal as possible without overloading the amplifiers. The bandwidth is the differ-

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ence between the upper and lower frequency at the given attenuation. The above test shall meet the requirements of 3.12.2.

4.19 Unwanted Sideband rejection. With a test signal applied to the IF input terminals of Converter, Single Sideband CV-157( )/URR, consisting of a carrier voltage of 1000 microvolts and a sideband voltage of 10,000 microvolts located at any frequency from 100 to 6000 cps above the carrier frequency, the audio output voltage of the lower channel shall be measured for compliance with 3.12.3. The same test shall be made with the test signal applied to the lower sideband, and the output voltage measured at the output of the upper sideband channel.

4.20 Hum level. With the carrier suppression control in the 0 db position, a signal at 455 kc modulated 30% at 400 cps shall be applied to the input terminals of Converter, Signal Sideband CV-157( )/URR. A 600 ohms resistor, and a General Radio Type 736-A Wave Analyzer, or equal, shall be connected to the A channel output terminals. The wave analyzer shall be tuned to 400 cps, and its indicated 400 cycles level noted. The modulation shall then be turned off, and the wave analyzer tuned to 60 and 120 cycles, and their respective levels noted. The square root of the sum of the squares of the 60 and 120 cycle levels shall be at least 50 db lower than the 400 cycle level. The above tests shall be repeated for channel B (see 3.13.4).

4.21 Single sideband noise and distortion. The intermodulation distortion shall be tested with a signal consisting of an IF carrier of 1000 microvolts at 455.0 kcs, and two sidebands each of 5000 microvolts between 455.1 and 461.0 kcs, or between 449.0 and 454.9 kcs: in either case differing from each other by not more than 500 cps. When this test signal is applied to the input terminals of Single Sideband Converter CV-157( ) URR, no intermodulation distortion component at either of the output terminals shall be of greater magnitude than 42 db below the output level of one of the two equal sideband voltages when the output level is set at 10 milliwatts (see 3.11.2).

4.22 Overall audio frequency response. With Single Sideband Converter CV-157 ( )/URR set for either upper or lower sideband reception, and the carrier suppression control set at 20 db, a single sideband signal of 10,000 microvolts together with a 1000 microvolt carrier shall be supplied to the converter input. The frequency of the sideband shall be varied above and below the carrier frequency by an amount between 100 cps and 6000 cps. A power output of 10 milliwatts shall be delivered to the specified resistance between 100 and 6000 cps, and the output voltage resulting from the sideband frequencies shall comply with the requirements of 3.13.3. On each side of the range 100-6000 cps, the output shall drop off smoothly without reverse curvature. When the lowpass filters are switched into the demodulator output circuits, audio frequency response shall conform to the requirements of 3.14.8.

4.23 AFC Pull-in Range. The Vernier is adjusted until the signal is tuned in. A Frequency Counter is connected to the junction of C81 and the coax-cable leading to R108. The frequency is noted. The Vernier is rotated clockwise until the indicated frequency is 50 cps higher. The carrier level meter should give no indication. The AFC Switch is now turned on. The meter should read mid-scale showing that the Converter retunes itself to the signal.

The AFC switch is turned off and the Vernier is rotated counter-clockwise until the frequency is 50 cps lower. The AFC switch is again turned on and the meter should indicate a tune-in condition.

To check hold-in over the range of ±2000 cps, with the AFC switch on, the Vernier is turned slowly, counterclockwise, until the Frequency Counter reads 2000 cps below the center frequency and then, clockwise, until the frequency counter reads 2000 cps above the center frequency. The meter should con-

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tinue to read mid-scale over the entire range. The pull-in range shall meet the requirements of 3.14.1.

4.24 AFC drift rate. With the carrier suppression control set at 20 db, a 1000 microvolt signal of any frequency within 2000 cps of the IF center frequency (determined by the center frequency of the carrier pass filter and the mixing oscillator frequency) shall be applied to the input terminals of Converter, Single Sideband CV–157( )/URR. The AFC circuit shall follow any rate of variation of the applied IF signal up to 10 cycles-per-second-per-second in either direction (provided that the maximum excursion does not exceed 2000 cps from the center frequency) without becoming unlocked (see 3.14.2).

4.25 Residual error. With the carrier suppression control set at 20 db, a 1000 microvolt carrier and a 10,000 microvolt sideband signal differing in frequency by 1000 cps, and both drifting at the rate of 10 cps per second, shall be impressed on the input terminals of Single Sideband Converter CV–157( )/URR set for single sideband operation with local carrier. The audio output signal shall be compared with a signal consisting of the beat note between the carrier and sideband input signals. The difference in frequency shall be determined for low difference frequencies by recording the difference frequency between the transmitted 1000 cps tone and the received 1000 cps tone. This difference shall not exceed 2½ cps (see 3.14.3).

4.26 Carrier noise-squelch circuit. With a carrier suppression control set at 20db, a test signal consisting of a 1000 microvolt carrier at 455 kc shall be applied to the input terminals of Single Sideband Converter CV–157 ( )/URR. After the AFC has locked-in to the center frequency, the applied signal shall be removed from the input terminals of the converter and reapplied at the end of a one minute period. The carrier level meter should indicate at this time that the AFC circuit has re-established control. Correct adjustment of the squelch circuit shall be demonstrated by having operation occur when two signals, one at 1000 microvolts, and the other 3 db lower, 5 cps removed in frequency, are simultaneously applied to the converter input, and are tuned to fall into the carrier channel (see 3.14.5).

4.27 Filter Test.

4.27.1 *Quartz Crystal Upper and Lower Sideband Filter Tests.* The upper and lower sideband filters shall be tested as a balanced structure between input and output terminations of 1560 ohms with a testing voltage not to exceed 0.5 volt, and with a room temperature of 75 ±5°F. Input terminals are #1 and #2. Output terminals are #3 and #4. The testing voltage of 0.5 volts shall be applied to input terminals numbered 1 and 2 of the filter test jig. The filters shall meet the insertion loss requirements specified in 3.17.1.

4.27.2 *Quartz Crystal Carrier Filter Test.* The carrier filter shall be tested as an unbalanced structure between input and output terminations of 50,000 ohms with a testing voltage not to exceed 2.0 volts, and with a room temperature of 75 ± 5F. This filter shall meet the insertion loss requirements specified in 3.17.1, under the following test conditions:

7.5 ±1 mmf shall be connected between terminal 3 and 6

16.0 ±1 mmf shall be connected between terminals 4 and 6

The shunt capacitance of each 50,000 ohm termination shall not exceed 5 mmf.

Input terminals are #1 and #5. Output terminals are #4 and #6.

4.28 Automatic gain control. The converter shall be connected to Radio Receiver R–390 ( )/URR, as detailed below. The following test shall be performed to determine compliance with 3.14.6 for each position of the AGC select switch.

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Squelch	— ON
AFC	— ON
Carrier Compensator	— 6
Carrier Selector	— LC
SB Selector	— B–USB
AGC Time	— Medium
AGC Selector	— REG
SB AGC	— As described below

Input signal to Converter. IF output from R–390 Radio Receiver. The R–390 is set up as follows:

RF Gain — Maximum; Bandwidth — 8 KC; Function Switch–AGC; BFO — OFF; Antenna Trimmer — tuned to give maximum carrier level meter reading; and Tuning Controls — set to 4.0 megacycles. Into the R–390, two signal generator outputs are fed. One generator supplies the carrier and is tuned to 4.000 MC at 5 microvolts level. The other generator supplies the sideband signal. It is tuned to 4.002 MC at 50 microvolts. At all settings of the two generators, the 20 db difference is always maintained. The Converter is tuned until the Carrier Level Meter gives a deflection. *Be careful that the Carrier signal, and not the sideband, is causing the deflection.*

There should be a 2,000 cps audio signal present in the Channel B output. A VT VM is connected across one of the 300 ohm load resistors. Volume control B–VC is adjusted until the meter reads .388 volts, or 0 DB (1 milliwatt across the full 6000 ohms). This is the reference level for the entire set of measurements. The SB AGC Control is set to give 0 LB output when the AGC Selector is in the USB position. The audio output levels, corresponding to the various input levels and positions of the AGC Selector switch (as indicated below) shall be recorded and shall meet the requirements of 3.14.6:

Receiver input (microvolts)		Converter audio output (DB) AGC selector switch position				
Carrier	Sideband	Rec	Car	LSB	USB	TSB
5	50	—	—	—	—	—
50	500	—	—	—	—	—
500	5000	—	—	—	—	—
5000	50000	—	—	—	—	—
50000	500000	—	—	—	—	—
100000	1,000000	—	—	—	—	—

4.29 Carrier level compensation control. For this test, the AFC control switch shall be in the "on" position and the carrier level compensation control shall be in the 0 db position. A 455 kc IF frequency shall be applied to Single Sideband Converter CV–157( )/URR to produce a half scale reading on the carrier meter. The carrier level compensation control shall then be rotated to the 10 db position, and the input voltage decreased until the carrier meter reading is the same as before. The ratio of this input voltage to that employed when the carrier level compensation control was at 0 db shall not vary from —10 db by more than plus or minus 1 db. The same test shall be performed with the carrier level compensation control in positions corresponding to 20 db and 30 db. The ratio of the final voltage to that employed when the carrier level compensation control was at the 0 db position shall not vary from —20 db or —30 db respectively by more than plus or minus 1 db (see 3.14.7).

4.30 Drift indicator. Mixing Oscillator inductor, L–1, is adjusted until a peak indication is obtained on the Carrier Level Meter. The Drift Indicator is then set to the –2000 cps position. The frequency of the audio signal output from Channel B, as indicated on the Frequency Counter, is recorded. This is repeated with the drift indicator set at —1500, —1000, —500 and 0 cps in turn. The Drift Indicator is then set at 500, 1000, 1500 and 2000 cps in turn. The SB Switch is changed to B–LSB in order that the audio Signal will be present in Channel B. The frequency is recorded. (If either of the audio frequencies, at both points, is not between 1800 and 2200 cps, the Range Compensator setting must be changed). Setting the Compensator to a lower number will reduce the

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frequency range of the Drift Indicator. With the SB switch set as appropriate and the Drift Indicator set to zero, the Vernier is set from —2000 to +2000 in 500 cps steps and the corresponding readings are recorded. With the AFC Switch in the ON position, the signal generator is slowly varied below, and then above, 455 kc. The frequencies at which the Drift Alarm Indicator lights and the AFC loses control is recorded. The frequency at the critical points may be determined by setting the AFC Switch to OFF and restoring the Drift Indicator to zero. The frequency of the audio signal present in the output channel is the required value. The requirements of 3.11.1 shall be met.

4.31 Monitoring Circuit. A 455 kc carrier at 1000 u v and a 10,000 u v sideband signal is applied to the input of the converter. The audio output of each channel is set to 1 m v by adjustment of A–VC and B–VC. An oscilloscope and a 600 ohm load are connected to the Monitor output, with the sideband signal set to give an audio output from Channel A. The Monitor and VU Select switches are set to A and VU Range switch is set to 0. The Monitor Gain Control is adjusted to the point where distortion of the Monitor output is noticeable on the oscilloscope. The power output across the 600 ohm load is recorded. The Monitor switch is then placed in all three of its positions. The requirements of 3.11.5 shall be met.

4.32 Altitude test. The equipment shall be placed in an altitude chamber at normal conditions of temperature, pressure, and humidity, and the performance measured. The pressure shall then be reduced to simulate 10,000 feet above sea level and stabilized for 2 hours. The performance of the equipment shall again be measured. The pressure shall then be lowered to stimulate 25,000 feet above sea level and stabilized for another 2 hour period. The pressure shall then be increased to sea level conditions and the performance measured again. The equipment shall have suffered no degradation in performance from that obtained prior to this test (see 3.18.2).

4.33 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine compliance with the physical interchangeability requirement of 3.20. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) External and internal dimensions of cases, covers, and insertable assemblies when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

4.34 Procurement inspection of preparation for delivery. The minimum inspection performed by the contractor shall be as specified in 4.34.1 and 4.34.2. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's records. In addition, the Government—at its discretion—will verify, by product inspection, the contractor's compliance with specified requirements. Such verification shall not exceed the amount of inspection required herein of the contractor. When MIL–STD–105 specifies actions by the Government, the Government may authorize the contractor to perform any of such actions except that responsibility for acceptance rests with the Government.

4.34.1 *Preservation and packaging.* Inspection of preservation and packaging shall be as specified in MIL–P–116. Classification of defects shall be as shown on table VI.

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TABLE VI. *Preservation, and packaging and marking thereof*

Major	Minor
1. Use of improper or defective material.	1. Wrong type of contact preservation.
2. Quantity in unit package not as specified.	2. Item not properly blocked or braced within the unit package to prevent movement.
3. Incorrect packaging method applied.	3. Packaging material damaged.
4. Cushioning or padding omitted.	4. Any item of marking information other than 7 through 9 omitted, incorrect, or illegible.
5. Cushioning inadequate for the physical and mechanical protection of the item.	
6. Unsealed, punctured, or improperly sealed barrier bag, wrap, or envelope.	
7. Stock number omitted, incorrect, or illegible.	
8. Nomenclature omitted, incorrect, or illegible.	
9. Marking of quantity of items in package omitted, incorrect or illegible.	
10. Different stock numbered items in the same unit package.	

4.34.2 *Packing and marking of exterior containers.* Packing and marking of exterior containers shall be given visual inspection for the defects listed below and to determine conformance with approved process sheet furnished by the contractor as required by the contract. This inspection shall conform to the sampling for general inspection levels by attributes in MIL–STD–105. Unless otherwise specified herein, normal inspection shall be used at the start of the contract. The AQL for minor defects shall be ten percent. Classification of defects shall be as shown on table VII.

TABLE VII. *Packing, and marking thereof*

Major	Minor
1. Use of improper or defective material.	1. Unsealed carton.
2. Quantity in pack not as specified.	2. Defective taping or sealing of carton.
3. Gross weight in excess of specified amount.	3. Any other box defect which may be considered minor by definition of MIL–STD–105.
4. Box closure not as specified.	4. Any item of required marking information other than 10 through 15 listed under major defects omitted, incorrect, or illegible.
5. Type, grade, class and style of the shipping container not as specified.	
6. Strapping inadequate or incorrectly applied.	
7. Items not adequately blocked, braced or cushioned within the shipping container to prevent movement or damage.	
8. Shipping documents or packing list omitted.	
9. Stock number omitted, incorrect or illegible.	
10. Nomenclature omitted, incorrect or illegible.	
11. Marking of quantity of items in pack omitted, incorrect or illegible.	
12. Destination marking omitted, incorrect or illegible.	
13. Service designation (color marking) omitted.	
14. Specified special marking and labeling such as NAP labels, shipment digit marking, etc., omitted, incorrect or illegible.	
15. Overseas code marking omitted, incorrect or illegible.	

4.34.3 *Inspection lot.* A lot for visual inspection of the pack shall be all completed packs which are identical and will be submitted for acceptance at one time.

4.34.4 *Procedure in case of failure.* If an inspection lot fails, the contractor shall immediately investigate the cause of failure and shall report to the Government inspector

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the results thereof and details of the corrective action taken. If the contractor and Government inspector cannot agree on the effectiveness of the corrective action, the matter shall be referred to the contracting officer for resolution.

4.34.5 *Disposition of nonconforming product.* Disposition of nonconforming product (sample units and inspection lots) shall be in accordance with the requirements of MIL–STD–105 for disposition of defected product. When submitted for acceptance, such product shall be suitably tagged or identified by equivalent means to indicate the cause of failure and means employed to correct the fault. The required information shall be presented to the Government when the product is submitted and shall become the property of the Governmnt.

4.34.6 *Rough handling test (preparation for delivery).* When rough handling test is rquired by the contract, the test of 4.19 shall be performed and the requirements of 3.12.3 shall be met (see 6.1(e)(2)*c*).

4.35 **Power Supply.** The equipment tested with the following variable input sources shall meet the requirements of 3.11.5 and 3.15:

Input V	*CPS*
103.5	62
115	62
126.5	62
207	62
230	62
253	62
103.5	48
115	48
126.5	48
207	48
230	48
253	48

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level A.* Converter CV–157( ) URR shall be preserved and packaged in accordance with the procedures specified for the designated methods as prescribed in MIL–P–116 and described herein.

5.1.1.1 *Technical manual.* Each technical manual shall be packaged Method IC–3 as follows: Inclose each technical manual within a close-fitting bag, type 1, class b, conforming to MIL–B–117. Heat seal to effect closure.

5.1.1.2 *Tubes.* Remove all operating tubes over four inches in length not securely clamped in place and package together with spare tubes, individually, in accordance with the applicable provisions of MIL–P–75.

5.1.1.3 *Cords CD–135, CD–370 and CG–409E/U.* Each cord shall be packaged Method III as follows: Wind each cord into a coil of proportionate dimensions and tie at 3 places with cotton twine, type, 8 ply conforming to Specification T–T–871. Secure connectors to the coil with the ties.

5.1.1.4 *Consolidated package.* Consolidate the items packaged as specified in 5.1.1.2 and 5.1.1.3 together within a close-fitting, regular slotted style, class 2 W5c fiberboard box, conforming to Specification PPP–B–636. Fill all voids with flexible, single face, corrugated fiberboard type I, conforming to Specification LLL–F–291. Closure shall be in accordance with the appendix of the box specification.

5.1.1.5 *Converter CV–157( )/URR.* Each converter shall be packaged Method IC–5, as follows: Place the adjustment tools in space provided within cover. Place cover over converter and secure all fastenings. Cushion each converter on all surfaces with cells or pads or both, fabricated of fiberboard, having a minimum dry bursting strength of 275 pounds, and designed to absorb the shock of impact, encountered in handling and transit. Cells and pads shall have a water resistance equal to that of the box in which they are used. Place each cushioned converter within a close-fitting, regular slotted style, class 2,

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W5c, fiberboard box conforming to Specification PPP–B–636. Seal the box as specified in the appendix of the specification and in addition seal all joints and seams with water resistant, pressure-sensitive tape, conforming to Specification PPP–T–76.

5.1.2 *Level C.* Converter Single Sideband CV–157( )/URR shall be packaged in accordance with commercial practice and in a manner that will afford protection against corrosion, deterioration and physical damage during direct shipment to the first receiving activity.

5.2 Packing.

5.2.1 *Level A.* Converter CV–157( )/URR, together with consolidated package, shall be packed within a close-fitting, overseas type or class shipping container as follows: boxes, wood, nailed and lock corners, boxes, wood, wirewound and boxes fiberboard, wood cleated conforming to Specification PPP–B–621, PPP–B–585 or PPP–B–591, respectively. Box closure shall be in accordance with the applicable box specification or the appendix thereto.

5.2.1.1 *Metal strapping.* Shipping containers shall be strapped in conformance with the requirements of the appendix of the applicable container specification.

5.2.2 *Level A.* Converter CV–157( )/URR shall be packed as specified in 5.2.1, except that the boxes shall be domestic type or class.

5.2.3 *Level C.* Converter CV–157( )/URR shall be packed for shipment in a manner conforming to the requirements of Uniform Freight Classification No. 4 and Consolidated Freight Classification Rules 21 for rail shipment, National Motor Freight Classification No. A–3 and No. 14 for truck shipment, Parcel Post Regulations and the regulations of other carriers as applicable to the mode of transportation employed, at the lowest transportation rate.

5.3 Marking. Interior packages and exterior shipping containers shall be marked in accordance with the applicable provisions of MIL–STD–129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Type required.

(c) Level of packaging and level of packing required for shipment. (Level A, level B, or level C.)

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction inspection:

(1) Two samples of each item cited in section 1 are generally required so that lengthy environmental tests can be completed on one sample while complete performance measurements can be made on the second sample (see 3.22).

(2) Preproduction pack(s) as follows:

a. Makeup of pack(s).

b. Number of each kind of pack to be submitted.

c. Inspection to be performed thereon-including rough handling test, which will not be performed as procurement inspection.

d. Five copies of process sheet, using format shown on Drawing SC–C–33073 and accompanied by such drawings, sketches, and other data as be necessary to completely describe the procedure followed in fabrication of the preproduction pack(s) and to identify the items therein. These data should be submitted to the

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contracting officer with the preproduction pack.

e. A packaging technician from the procuring agency will evaluate preproduction pack(s).

f. Marking and shipping of samples.

g. Place of final inspection.

h. Technical literature required (see 3.23).

i. Quantity of tools and running spare parts required (see 3.23).

j. Submission of the statement of treatment referenced in 3.9 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.2 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: AN/PIQ-IW. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.3 Group C Inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (see 4.4.3.2).

6.4 Definition of inspection terms. The inspection terms used herein conform to MIL-STD-109, the following are abbreviated definitions selected from those given therein for immediate reference:

6.4.1 *Inspection.* Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.4.1.1 *Examination.* Examination consists of simple, generally nondestructive determinations of compliance, without use of special testing equipment.

6.4.1.2 *Testing.* Testing consists of determinations of compliance, using technical means.

6.5 Verification inspection (see 4.4). Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.6 Color. The color chip furnished by the procuring agency will be color chip No. 2430, and may be obtained upon request to Commanding Officer, U. S. Army Electronics Command, 225 South Eighteenth Street, Philadelphia, 3, Pa.

Custodian:

Army—EL

Preparing activity:

Army—EL

Project No. 5820-A328

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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